Molecular Modeling in Process Engineering

2025/2026

Project 4

1) *Quantum Chemical based Calculation of energies and energy changes: effect of basis set*

The purpose is to compute the impact of basis set on the determination of energies at three different levels of theory.
It is requested to determine the energy change for the reactions:

$H_2 \rightarrow 2H$
$CH_4 \rightarrow CH_3 + H$
$CH_3NH_2 \rightarrow CH_3 + NH_2$

At the following levels of theory:

HF, DFT (B3LYP and other functionals of your choice, such M06-2X, B2PLYP-D3, etc), MP2, and CCSD(T).

As basis set you are requested to use at least the Pople and correlation consistent (cc) families of basis sets.
When performing the calculations:
- try to use some of the calculations specifying the basis set in the input.
- try to impose the symmetry, at least for methane and methyl, when performing the calculations, in order to speed up the calculations. Take note of the symmetry group identified by the code.
- calculate by hand the number of basis and gaussian functions for at least three different basis sets for three different systems.
- Try to find out appropriate expression to determine energies at what is called the 'basis set limit', or complete basis set.

Remember to include the zero point energy corrections if you want to determine the correct bond energy. The experimental bond energy is 103.2 kcal/mol for H2 and 103.3 for CH4. Try to check these numbers and to determine the proper value for methylamine using AtCT data (https://atct.anl.gov/).

2) *Estimation of the equilibrium constant from the molecular partition function*

Estimation of the equilibrium constant from the molecular partition function
The purpose is to learn how to compute partition functions and an equilibrium constant from the data determined using electronic structure calculations. It is requested to determine for all species involved in the reaction, in the 300-2000 K temperature regime the following quantities:

- translational partition function
- vibrational partition function
- rotational partition function
- electronic partition function

And calculate the equilibrium constant of the following reaction: $CH_4 \leftrightarrow CH_3 + H$

All the necessary data can be evaluated through optimization following by frequency calculation for each considered molecule. Useful equations, together with their derivation, are reported in chapter 4 of the ACK lecture notes, that you can find on BEEP. It is requested to perform all the calculations using an excel worksheet (to validate the implemented equations through comparison with the data determined using quantum chemical calculations).